



**Project
Management
Institute.**



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PROJECT MANAGEMENT

IREC supports your career journey.

Project Cost Components.

- The Project Cost Management includes the processes required to carry out – **economic evaluation which initiates the project, estimating, controlling, analyzing, reporting and forecasting costs as well as taking the necessary corrective action.**
- After defining the **project scope** and determining the project activities, **each project deliverable and related activities will have an associated cost.**

PROJECT COST MANAGEMENT.

It is done through four processes, namely:

1. **Plan cost management.**
2. **Estimate activity costs.**
3. **Determine budget.**
4. **Control costs.**



1. Plan Cost Management Process.

- It involves defining processes of how to estimate costs, develop the budget and control the expenses throughout the project.
- A cost management plan is the primary output of this process.

Plan Cost Management – Inputs.

- **Project charter** – contains pre-approved financial resources that are used in developing a detailed budget.
- **Project management plan** – schedule management plan and risk management plan that contain controls that impact cost estimation.
- **Enterprise environmental factors** – resources that are available on the local, regional market and their prices. The exchange rates between currencies and levels of inflation.
- **Organizational process assets** – these could include financial control procedures, historical information, financial databases and budgeting policies/processes.

Plan Cost Management – techniques/tools.

- **Expert judgement** – expertise should be considered from individuals and groups with knowledge in industry best practices, cost estimating and earned value management.
- **Data analysis** – analyzing different ways of acquiring resources such as make or buy or lease as well as funding options such as self-funding, debt, equity or grants.
- **Meetings** – convening meetings to discuss and agree on the most appropriate approach project funding involving the sponsor, project manager, selected project team members and selected stakeholders.

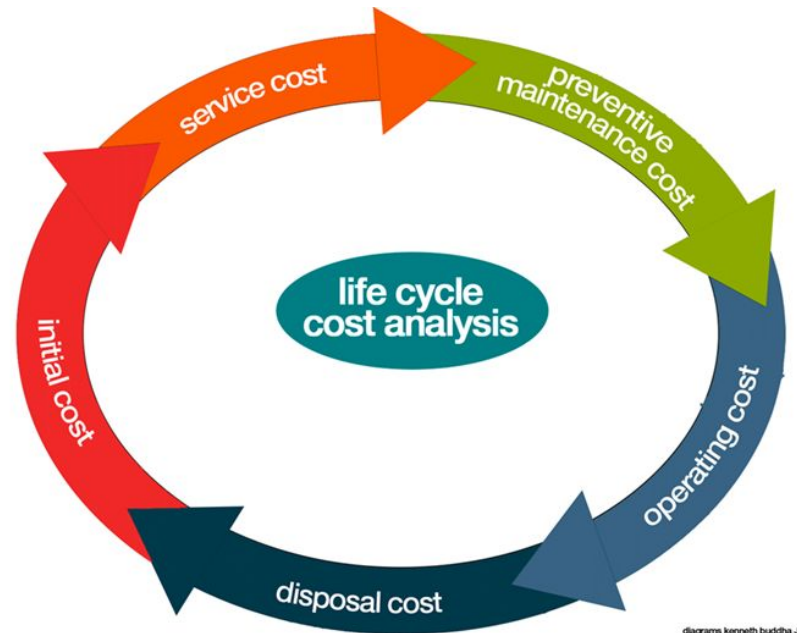
Contents of the Cost Management Plan:



1. Units of measure – currencies or effort e.g cost/staff-hour.
2. Levels of Precision/accuracy is important, especially if you are rounding numbers. It is critical that your figures be as precise and accurate as possible.
3. Control thresholds and allowable cost variations.
4. Rules of performance measurement – earned value analysis.
5. Reporting formats for monitoring/control purposes.
6. Any other costing details such as strategic funding choices.

The Cost Management Plan also involves:

Project Life Cycle Cost– Everything that is produced has a lifetime also called '*cradle to grave costing*'. It covers the total cost of ownership from purchasing or creation through operations and maintenance and finally to disposal of the product, service or result of the project.



Value Engineering.

- **Value engineering** also known as **Value Analysis** is used to analyze a service, system, or product to determine the best way to execute important activities efficiently (*lower cost/shorter time*).
- However, this does NOT mean reducing scope or decreasing quality to deliver project work with a lower cost.
- **Value engineering** is performing the same work with a less costly method. *For instance, if you need a bulldozer for your construction project only for 9 months, most probably, leasing the bulldozer will be less costly than buying a bulldozer if you won't use it in your future project works.*

2. Estimate Activity Costs process.



- **Cost Estimation Process** develops an approximation of the cost of resources needed to complete project work.”
- Based on the activity estimates, the overall project budget is estimated.
- **Resource costs include fixed and variable costs:**
 1. Fixed costs are such as office rent. Regardless of how many project resources will work, you need an office, and this cost will be in your project budget.
 2. Variable costs: this cost varies with amount of work/production e.g. cost of **material costs and other consumables**, which vary according to the quantities of resources.

Cost Estimating.

Developing an approximation of the cost for each activity in a project.

A word cloud containing the following terms: IT, Labor, Materials, Facilities, Equipment, Services, and Reserve. The words are arranged in a roughly triangular shape, with 'IT' at the top and 'Reserve' at the bottom. Each word is a different color and font size, with 'Facilities' and 'Equipment' being the largest.

IT
Labor
Materials
Facilities
Equipment
Services
Reserve

Estimate Costs – General Guidelines:

- Gather estimates for individual work packages.
- Check with the resource supplier to validate assumptions.
- Choose a suitable estimating technique according to context.
- Look for alternative costing options.
- Determine which units of measure to use.
- Consider impact of risks on cost.
- Ensure that cost estimates are assigned to the right account.
- Ensure estimates include resource costs, level of estimate, and a list of assumptions.

ANALOGOUS ESTIMATING.	PARAMETRIC ESTIMATING.	THREE POINT ESTIMATING.	BOTTOM-UP ESTIMATING.
Can be used for cost, duration and resource estimating.	Can be used for cost, duration and resource estimating.	Can be used for cost, duration but not resource estimating.	Can be used for cost, duration and resource estimating.
Uses historical information and expert judgement for estimation.	Uses historical information, mathematical models and parameters.	Uses average of three possible estimates to get final estimate. $E = (O + ML + P)/3$ or $E = (O + 4ML + P)/6$.	Estimate is done for the lowest level and aggregated upwards to get the final estimate.
Least accurate.	More accurate than analogous.	More accurate than analogous and parametric.	Most accurate technique.
Least time-consuming technique.	More time-consuming than analogous.	More time-consuming than analogous and parametric.	Most time-consuming technique.

Bottom-up estimating technique.

- Bottom-up estimating is **the most accurate method** to determine the definitive cost estimate of a project.
- It involves estimating for the most granular component level of a project – work package or an activity. The results of these granular estimations are then rolled up to calculate the estimate for the whole project.
- It requires more effort and **an established work breakdown structure (WBS)**. The project must have been broken down to the work package and activities level before applying this technique.
- The estimates are accurate also because they are **produced by the team members responsible for the work packages/activities/tasks**.

Cost of quality.

- Although ‘**Cost of quality**’ is not a generic estimation technique on its own it involves **determining the cost of conformance with certain quality standards compared with the cost of non-conformance and the long-term effects of short-term cost savings.**

Basis of estimates.

- The basis of estimates requires attachments which include the *estimate deliverable checklist*, *reference documents*, *benchmarking reports*, *risk analysis* and *reconciliation reports* among many others.
- It is important to take note that the basis of estimates is concise, factual and should describe techniques used to create the cost estimate.

3. Determine Budget Process.

- This is the process of aggregating the estimated costs of individual activities or work packages to establish an **authorized cost baseline**.
- The key benefit of this process is that it **determines the cost baseline** against which project performance can be monitored and controlled.



Determine a Budget – General Guidelines:

- **Review:**
 - Cost management plan
 - Resource management plan
 - Scope baseline for project scope statement, WBS, and WBS dictionary
 - Risk register to consider any risks that may impact cost estimation
 - EEFs and OPAs
- **Check the project schedule** for type, quantity, and duration of resources.
- Use appropriate tools and techniques.
- Document the project budget, creating a **cost baseline**.
- Understand project funding requirements or cash flow to enable the project.
- Update project documents, as needed.



Project Governance.

- ✓ Budget management is a critical project oversight and within the purview of project governance.
- ✓ Deviations in budget, scope, schedule, resources or quality, will impact the project.
- ✓ Project governance tells you whom these issues would impact and how to deal with them.
- ✓ Tailor cost estimation approach to phases of the project life cycle.

Budgeting process.

- After all activity cost estimates are done, they are **aggregated** (*summed up*) during this process and the overall project budget is determined.
- **Reserve Analysis:** Reserves are also determined namely, activity **contingency reserves** and the **project management reserve** to accommodate any risks if they occur during the project.
- **Cost Baseline** is the authorized time-phased budget for the project on which project cost performance is measured against.
- **Cost Baseline** = Project Cost Estimates + Contingency Reserves

Contingency Reserve

- For managing **Known-Unknowns**
- **Part** of Performance Measurement Baseline
- Project Manager has **full authority** over these reserves.
- These reserves can be used at **PM's discretion**.
- These reserves can be allocated at the **activity level**.
- These reserves are **calculated** using various techniques such as decision tree analysis, Monte Carlo simulation, EMV etc.

Management Reserve

- For managing **Unknown-Unknowns**
- **Not part** of Performance Measurement Baseline.
- Project Management has **no authority** over these reserves.
- Usage of these reserves requires a **formal request**, which should be approved by the higher management.
- These reserves can be allocated only at the **project level**.
- These reserves are not estimated, but are **fixed** according to organizational norms.

Funding Limit Reconciliation.



DEFINITION

The process of comparing the planned expenditure of project funds against any limits on the commitment of funds for the project to identify any variances between the funding limits and the planned expenditures.

Funding Limit Reconciliation.

Keep in mind:

- ✓ Most budgets assume steady incoming and outgoing flows.
- ✓ Large, sporadic expenditures are usually incompatible with organizational operations.
- ✓ Funding limits help regulate the outgoing capital flow to protect against overspending.



Establish a Cost Baseline – General Guidelines:

- Gather inputs to establish the baseline e.g. WBS, project schedule, cost estimates, and risk management plan.
- Assign work to dates on project schedule and allocate funds for each activity or work package for assigned time period.
- Consider a contingency reserve to cover expenses associated with risks.
- Total the costs for each time period, then plot these on a chart to create an S-curve of the baseline.
- Publish and distribute the cost baseline to the appropriate project stakeholders.



Budget Challenges.

- ✓ Ideally, a budget is set during project planning and does not change.
- ✓ However, the following changes can pose a challenge:
 - New/changed project requirements.
 - New risks, or changes to the probabilities or impacts of existing risks.
 - Changes to cost estimates resulting from economic factors, procurement contract modifications, resource costs, etc.

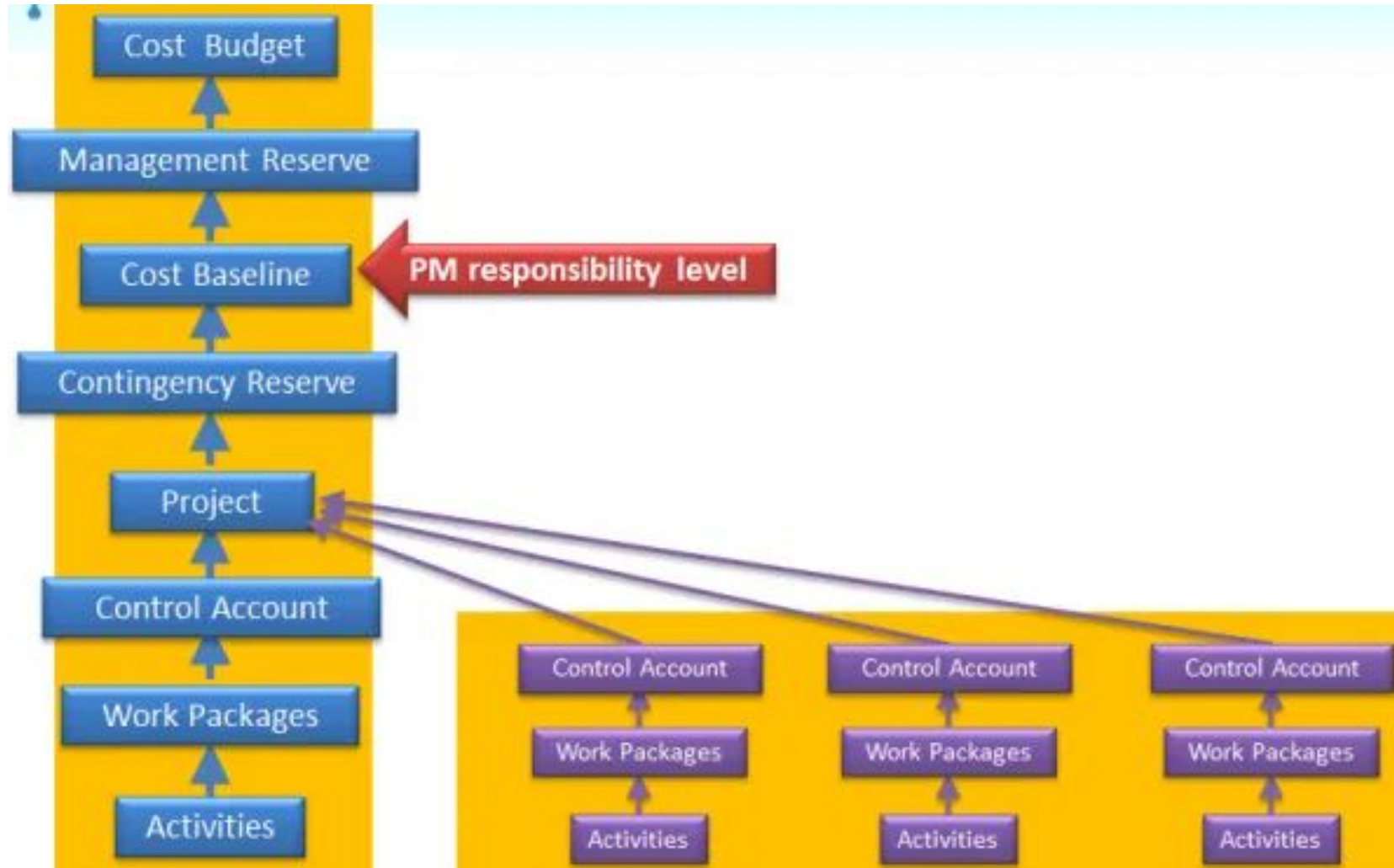
Cost Aggregation.



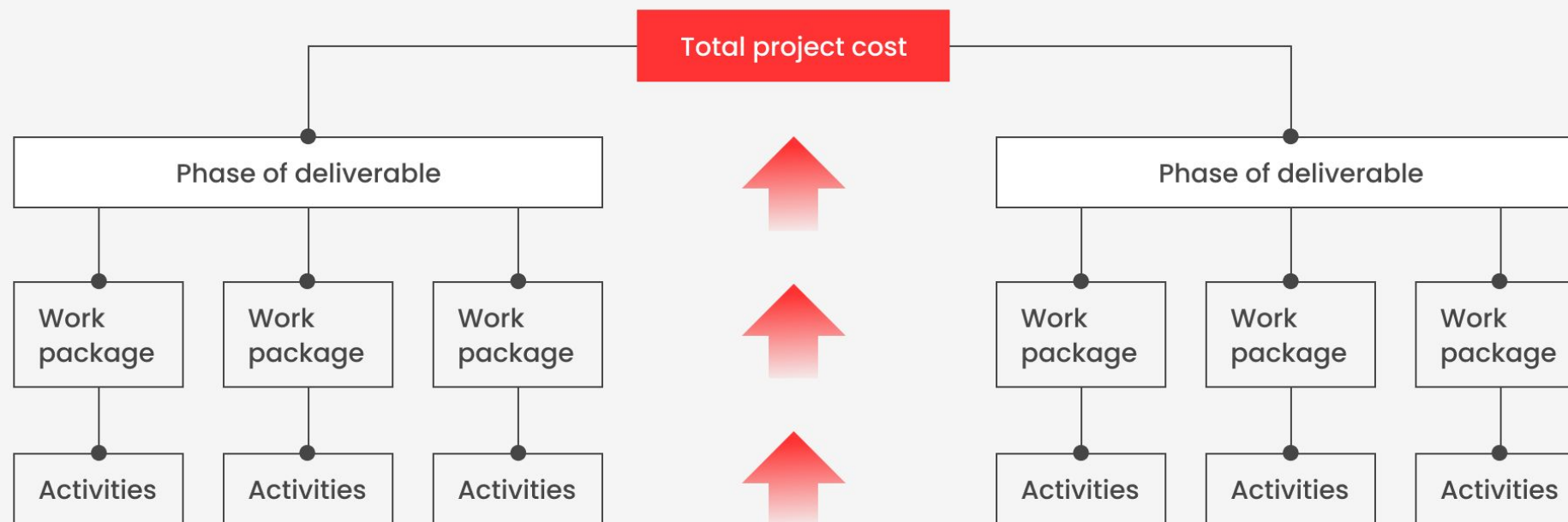
DEFINITION

Cost aggregation is the process of adding up the estimated costs of individual work packages or activities within a project to determine the total project budget.

Cost aggregation



Bottom-up estimation technique



Cost Baseline.



DEFINITION

The cost baseline is the approved version of the time-phased project budget, excluding any management reserves.

4. Control costs process.

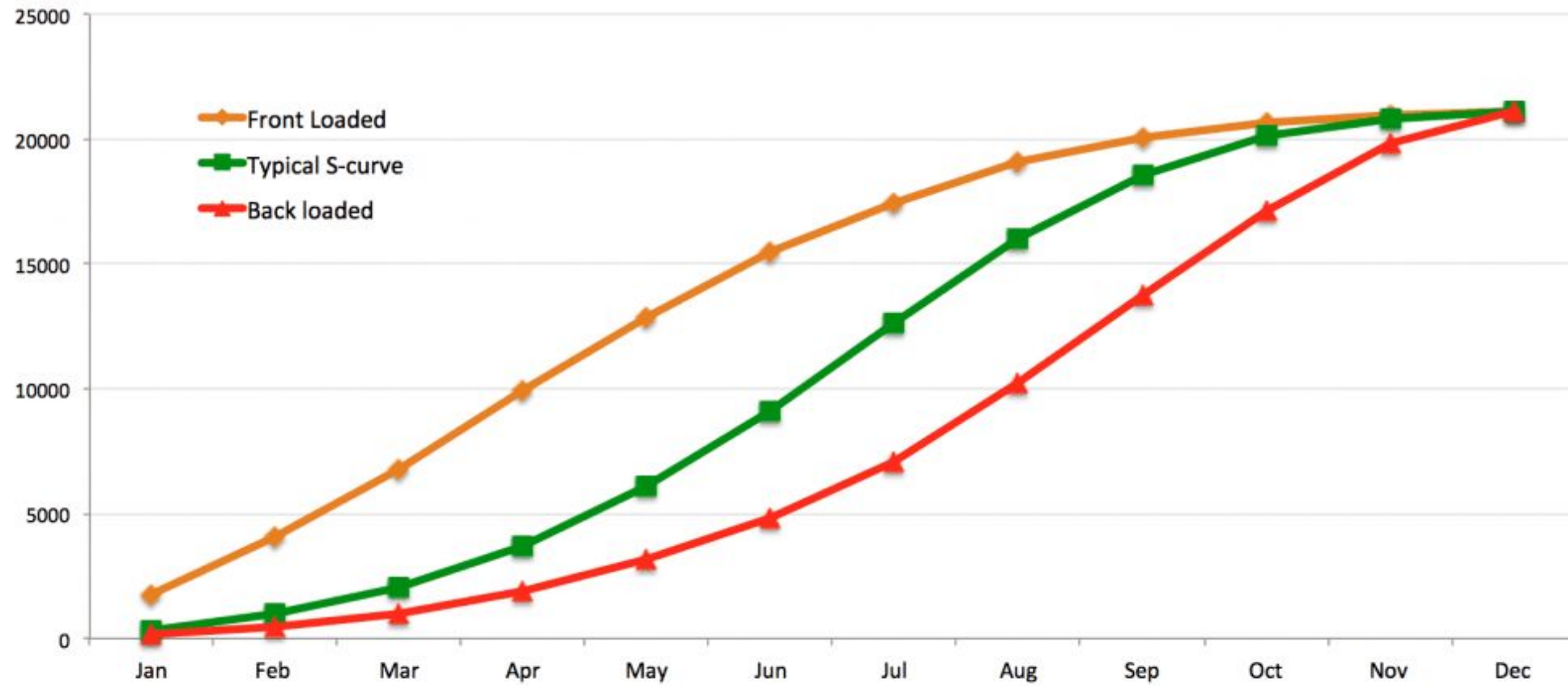
- This process aims to control the project expenses in order to complete the project within the determined budget using:
 - **Progress reporting** based on the previous expenses and tracking, whether the remaining activities will be completed with the remaining budget. Progress reports are submitted to relevant stakeholders.
 - **Earned Value Analysis** to determine whether the project is over or under budget. The analysis is critical in evaluating whether the project will meet the schedule and cost targets.

The role of the PM in cost control.

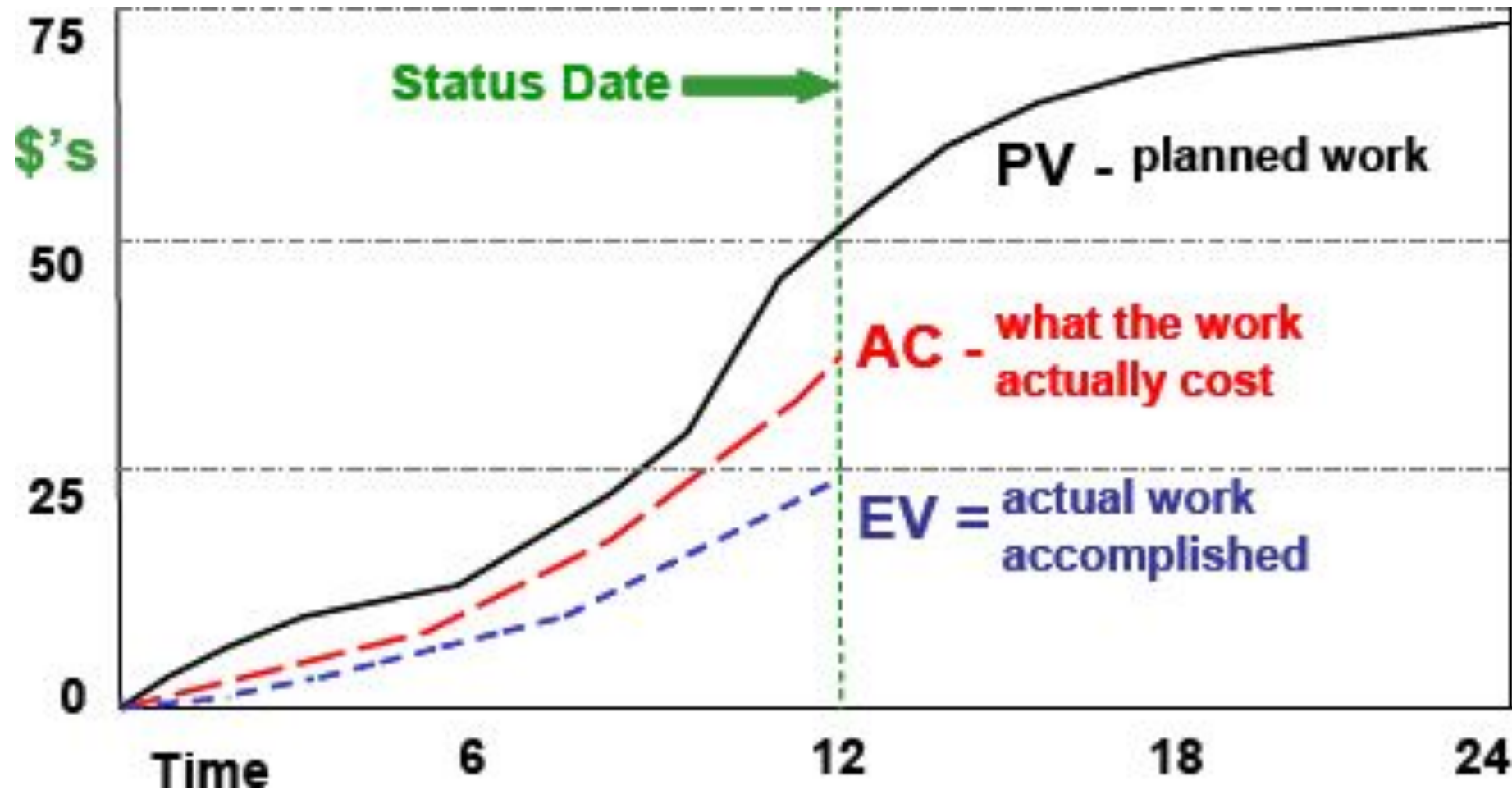
- The project manager *determines progress, measures variance, predicts outcomes, reports status, manages change and judge's outcomes.*
- **Earned Value (EV)** is a formal quantitative control mechanism used to detect and predict problems early, minimize speculation, evaluate status, and report progress. **It utilizes the planned estimates, earned value and actual costs obtained from work performance data.**

A time-phased budget

S-curve Shapes



Earned Value Management.



Earned Value
Analysis derives
information
from EV, PV and
AC.

- **Earned Value (EV)** is the value of the work actually completed to date.
- **Planned Value (PV)** is the value that should have been earned to date as per the schedule.
- **Actual Cost (AC)** is the amount spent on the project to date.

Earned Value Management

Project information.

- Project duration = 24 months.
 - Total Budget = 75,000.
- Reporting time = End of month twelve.
 - **Planned Value (PV) = US\$ 50,000.**
 - **Earned Value (EV) = US\$ 25,000.**
 - **Actual Cost (AC) = US\$ 35,000.**

Cost Variance (CV).

- **Cost Variance (CV)** is how much the cost is varying from the budget, based on what we know today.
- Formula: $CV = EV - AC$
- $CV = 25,000 - 35,000 = -10,000$
- CV – how much has been spent Vs work completed.

- If the CV is positive, its 'under budget' (good)
- If the CV is zero, the project is 'on budget' (ok).
- If the CV is negative, its 'over budget' (bad).
- Ex 1. If the CV for your project is +\$4,000, what does it mean?
- Ex 2. If the CV for your project is -\$4,000, what does it mean?

Schedule Variance (SV).

- **Schedule Variance (SV)** is how much the project is varying from the schedule as of today.
- Formula: $SV = EV - PV$
- $SV = 25,000 - 50,000 = -25,000$
- If the SV is positive, its ahead of schedule (good)
- If the SV is zero, the project is on schedule (ok)
- If the SV is negative, the project is behind schedule (bad).
- Ex 1. If the SV for your project is $-\$2,000$, what does it mean?
- It means you planned to complete $\$2,000$ more work than you did, as of today.

Performance indices.

- There are two Earned Value indices used in project management. The first is the **Cost Performance Index (CPI)**. The CPI is the ratio of budgeted to actual costs for work performed. Specifically, one can determine the relationship between estimated and actual costs. The second is the **Schedule Performance Index (SPI)**. The SPI is the ratio of work performed to work scheduled. The SPI is an efficiency rating for work accomplished up to the time of the measurement

Cost Performance Index (CPI)

- **Cost performance index (CPI)** is how well the project is performing from a cost standpoint as of TODAY.
- Formula: $CPI = EV/AC$
- $CPI = 25,000/35,000 = 0.7$
- For every Dollar spent we get 0.7 in value of work done.
- If the $CPI > 1$, the project is under budget (good).
- If the $CPI = 1$, the project is on budget (ok).
- If $CPI < 1$, the project is over budget (bad).
- Ex 1. If the CPI is 1.5, what does it mean?
- Ex 2. If the CPI is 0.5, what does it mean?

Schedule performance index (SPI)

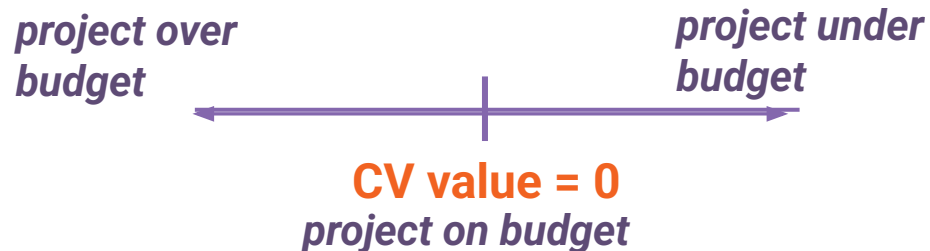
- **Schedule performance index (SPI)** is how quickly the project is progressing, compared to the schedule, as of TODAY.
- Formula: $SPI = EV/PV$
- $SPI = 25,000/50,000 = 0.5$
- We are therefore progressing at a rate which is a half of what we had planned.
- If the $SPI > 1$, the project is ahead of schedule (good).
- If the $SPI = 1$, the project is on schedule (ok).
- If $SPI < 1$, the project is behind schedule (bad).
- Ex 1. If the SPI is 1.8, what does it mean?
- Ex 2. If the SPI is 0.8, what does it mean?

EVM Measures for Cost Control.

Is the project on budget?

Calculate **cost variance (CV)** to find the current amount of budget deficit/surplus

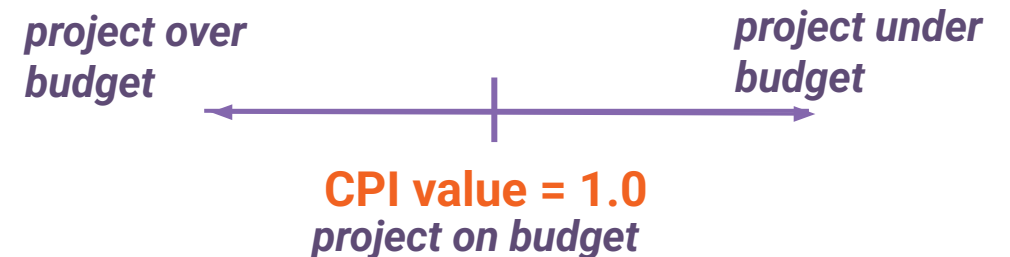
$$CV = EV - AC$$



How efficient is my project?

Calculate **cost performance index (CPI)** to measure the cost efficiency of budgeted resources

$$(CPI = EV / AC)$$

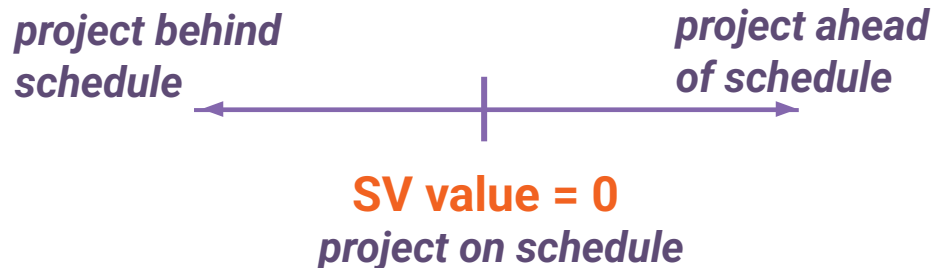


EVM Measures for Schedule Control.

Is the project progressing on schedule?

Schedule variance *measures performance – by calculating the difference between EV and PV*

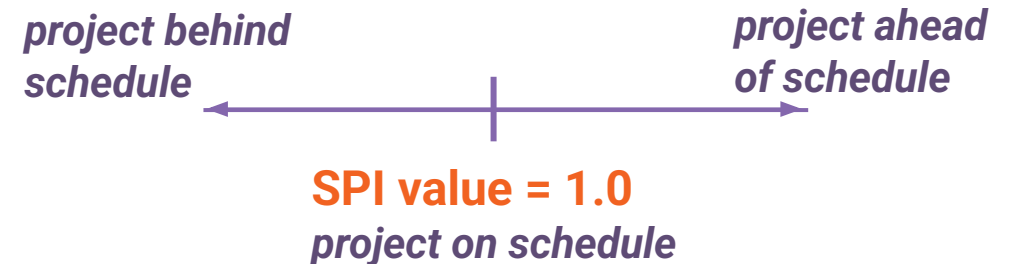
$$SV = EV - PV$$



How efficiently is the team working?

Schedule performance index *measures efficiency by calculating the ratio of EV to PV*

$$SPI = EV / PV$$



Estimate at Completion (EAC).

- **Estimate at Completion (EAC)** is how much we estimate the project will cost in total, based on what we know today.
- Formula: $EAC = BAC / CPI$
- $EAC = 75,000 / 0.7 = 107,142$
- BAC=Initial Budget at Completion.

- If the $EAC > BAC$, we expect the project to overrun the budget (bad).
- If the $EAC = BAC$, we expect the project to be complete on budget (ok).
- If $EAC < BAC$, we expect the project to underrun the budget (good).
- Ex. If EAC is \$7,000 and BAC is \$9,000, what does it mean?

Estimate at completion – EAC.

1. **EAC = AC + Bottom-up ETC**

This formula is used when the original estimation is fundamentally flawed. It calculates the actual plus new the estimates for the remaining work.

2. **EAC =BAC/CPI**

This formula is used when the existing circumstances are expected to be the same until the end (the cause of the deviation is typical).

3. **EAC = AC + (BAC – EV)**

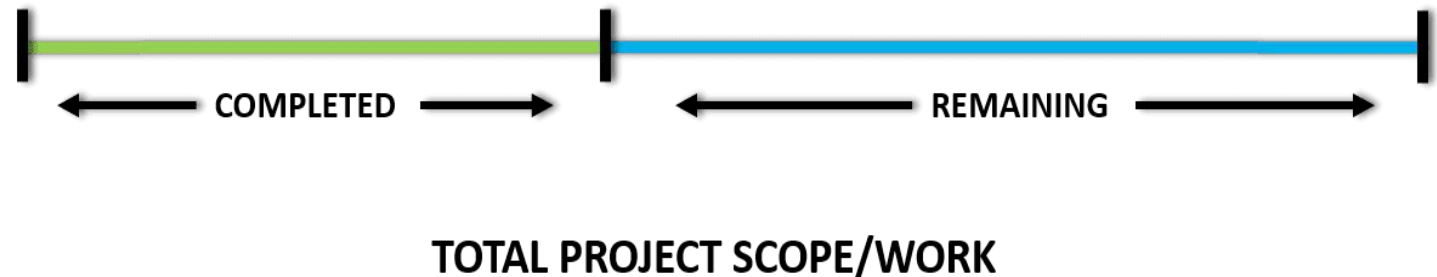
This formula is used when the current deviation with the original estimation is thought to be different in the future (the cause of the deviation is atypical). It is generally AC plus the remaining value of the work to perform.

4. **EAC = AC + [BAC – EV / (CPI x SPI)]**

This formula is used to calculate actual to date plus the remaining budget changed based on the performance. It is used when we believe the current ratio (CPI and SPI)is typical as planned.

Estimate to Complete (ETC)

- **Estimate To Complete** is an estimation of funds required to complete the remaining work in a project *i.e over and above what we have spent so far, in order to complete the project.*
- Formula: $ETC = EAC - AC$
- $ETC = 107,142 - 35,000 = 72,142$



Variance at Completion (VAC).

- **Variance at Completion (VAC)** is how much we estimate the project will overrun or underrun the budget, based on what we know today.
- Formula: $VAC = BAC - EAC$
- $VAC = 75,000 - 107,142 = -32,142$
- If the $VAC > 0$, we expect the project will underrun the budget (good)
- If the $VAC = 0$, we expect the project will complete on budget (ok).
- If the $VAC < 0$, we expect the project will overrun the budget (bad).
- Ex 1. If EAC is \$7,000 and VAC is \$2,000, what does it mean?

To Complete Performance Index (TCPI).

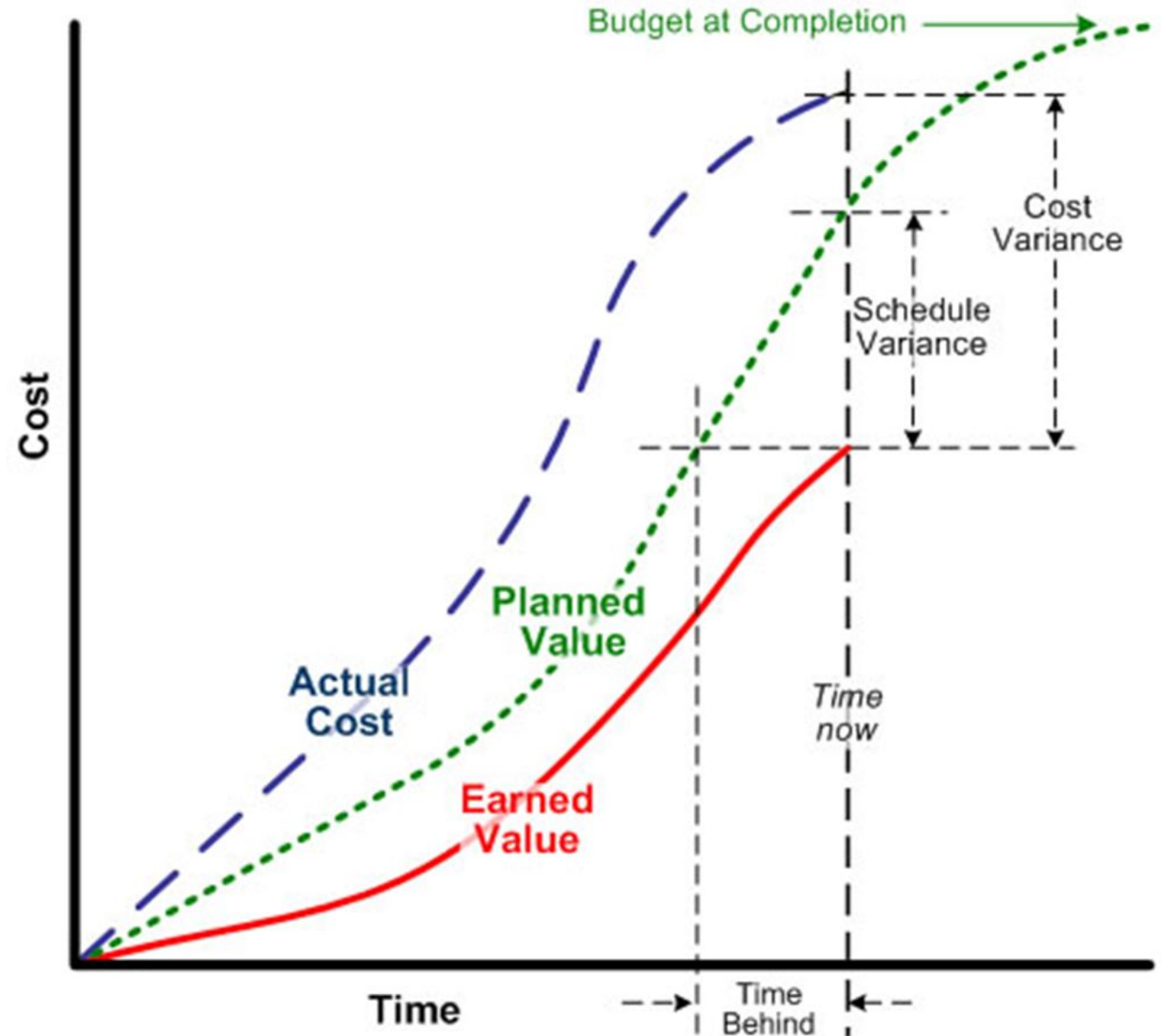
- **TCPI** is the cost performance index required to complete the project within management mandates.

- Formula:

$$TCPI = [BAC - EV] / [BAC - AC]$$

- If the $TCPI > 1$, we must achieve a better CPI than so far on the project (hard)
- If the $TCPI = 1$, we must achieve same CPI, at a minimum, as so far on the project (ok).
- If the $TCPI < 1$, we do not need to perform as well as the performance has been to date on the project (good).

Earned Value Analysis.



PRACTICE QUESTIONS

1. In a medium-sized healthcare project with an initial budget of \$ 800,000, the following information has been reported. Planned value = \$240,000, Earned value = \$140,000 and the Actual cost = \$180,000. Calculate the following:
- Cost Variance and Schedule Variance.
 - Cost Performance Index and Schedule Performance index.
 - Estimate at Completion.
 - Estimate to Complete.
 - Variance at Completion
 - TCPI.

Answer



- a. Cost Variance (CV) = $EV - AC = \$140,000 - \$180,000 = -\$40,000$
Schedule Variance (SV) = $EV - PV = \$140,000 - \$240,000 = -\$100,000$
- b. Cost Performance Index (CPI) = $EV / AC = \$140,000 / \$180,000 = 0.778$
Schedule Performance Index (SPI) = $EV / PV = \$140,000 / \$240,000 = 0.583$
- c. Estimate at Completion (EAC) = $BAC / CPI = \$800,000 / 0.778 = \$1,028,277.63$
- d. Estimate to Complete (ETC) = $EAC - AC = \$1,028,277.63 - \$180,000 = \$848,277.6$
- e. Variance at Completion (VAC) = $BAC - EAC = 800,000 - \$1,028,277.63 = -\$228,277.63$.
- f. TCPI = $[BAC - EV] / [BAC - AC] = [\$800,000 - \$140,000] / [\$800,000 - \$180,000] = 1.065$

PRACTICE QUESTION.



Qn.2 Calculate the CPI and SPI for each month.

Month	Planned Value (PV)	Earned Value (EV)	Actual Cost (AC)
1	20,000	18,000	22,000
2	40,000	35,000	40,000
3	60,000	55,000	58,000

Answer

Cost Performance Index (CPI): $CPI = EV / AC$

- Month 1: $\$18,000 / \$22,000 = 0.818$ (CPI < 1 indicates inefficiency)
- Month 2: $\$35,000 / \$40,000 = 0.875$ (CPI < 1 indicates inefficiency)
- Month 3: $\$55,000 / \$58,000 = 0.948$ (CPI < 1 indicates inefficiency)

Schedule Performance Index (SPI): $SPI = EV / PV$

- Month 1: $\$18,000 / \$20,000 = 0.90$ (SPI < 1 indicates delay)
- Month 2: $\$35,000 / \$40,000 = 0.875$ (SPI < 1 indicates delay)
- Month 3: $\$55,000 / \$60,000 = 0.917$ (SPI < 1 indicates delay)

PRACTICE QUESTION.



3. Attempt the following questions:

- A. What is EV, if your CPI is 1.1, your SPI is 0.92 and your AC is \$10,000?
- B. What is SPI and AC, if your CV is \$10,000, your SV is -\$3,000, and your PV is \$100,000?

Answer

A. $CPI = EV/AC$, meaning that $EV = CPI \times AC = 1.1 \times 10,000$.

Therefore, **EV = \$11,000**

B. $SV = EV - PV$, therefore $EV = SV + PV = -3,000 + 100,000 = \$97,000$

$SPI = EV/PV = 97,000/100,000 = \mathbf{0.97}$;

$CV = EV - AC$, therefore $AC = EV - CV = \$97,000 - \$10,000 = \mathbf{\$87,000}$

PRACTICE QUESTION

4. You are working on a project to set up a call center for your client. The duration of the project is two years. One year has passed, however, only 40% of the work has been completed against the schedule, which says that by now 50% of the work should have been completed. To date you have spent 60% of the budget. What is the cost performance index of the project?

- A. 1.25
- B. 1.5
- C. 0.8
- D. 0.67

Answer - D

In the question, you have not been provided with the budget, therefore let us assume that the project budget is 'X'.

- Earned value = 40% of budget = $0.4X$.
- Actual cost = 60% of budget = $0.6X$.

$$\text{CPI} = \text{EV/AC} = 0.4X/0.6X = 0.67.$$

PRACTICE QUESTION



5. Your project has a budget of \$900,000 and is running well. In the latest earned value report, the team reported that the $CPI=1.1$, the $SPI=0.9$, and the $PV=\$600,000$. You want to know, from this point on, how much more the project will cost but could not find it in the report. What will be the estimate to complete, or ETC in this case?

Answer



Information provided:

- BAC = \$900,000.
- CPI = 1.1; SPI = 0.9.
- PV = \$600,000.

What are the value for ETC?

The formula is $ETC = EAC - AC$

$$EAC = BAC/CPI = 900,000/1.1 = \$818,181$$

$$SPI = EV/PV \text{ implying that } EV = SPI \times PV = 0.9 \times 600,000 = \$540,000$$

$$CPI = EV/AC \text{ implying that } AC = EV/CPI = 540,000/1.1 = \$490,909$$

$$\text{Therefore, } \mathbf{ETC = 818,181 - 490,909 = \$327,272}$$

PRACTICE QUESTION.

6. If earned value (EV) = 350, actual cost (AC) = 400, and planned value (PV) = 325, what is the cost variance?

- A. 350
- B. -75
- C. 400
- D. -50

Answer



D. The formula is $CV = EV - AC$. Therefore, $CV = 350 - 400$, or $CV = -50$

PRACTICE QUESTIONS

7. If a project has a CPI of 0.95 and SPI of 1.01, this indicates:

- A. The project is progressing slower and costing more than planned.
- B. The project is progressing slower and costing less than planned.
- C. The project is progressing faster and costing more than planned.
- D. The project is progressing faster and costing less than planned.**

Answer



C. The project is progressing faster and costing more than planned.

PRACTICE QUESTIONS

8. If budget at completion = \$500, estimate to complete = \$400, earned value = \$100, and actual cost = \$100, what is the estimate at completion?

- A. 0
- B. \$150
- C. \$350
- D. \$500

Answer

D. $EAC = BAC / CPI$

- $CPI = EV / AC = \$100 / \$100 = 1$
- $BAC = \$500$



PRACTICE EXAMPLES

9. You are over budget and it seems difficult for you to complete with your current budget which is 250,00 USD. Considering your current cost performance will be the same as the future performance, you calculate your new budget as 300,000 USD. To date, you have spent 100,000 USD. What is the earned value?

- A. 150,000 USD.
- B. 120,000 USD.
- C. 83,000 USD.
- D. 100,000 USD.

Answer - C

Given in the question:

BAC = 250,000 USD

EAC = 300,000 USD.

AC = 100,000 USD.

$EAC = BAC / CPI$;

Therefore, $CPI = BAC / EAC = 250,000 / 300,000 = 0.83$.

$CPI = EV / AC$;

$EV = CPI \times AC = 0.83 \times 100,000 = 83,000 \text{ USD}$

PRACTICE QUESTION

10. You are managing a multi-year project whose budget is 500,000 USD. The cost variance is -45,000 USD, the actual cost is 200,000 USD, and the planned value is 140,000 USD. Based on the above data which statement is correct?

- A. The project is behind schedule.
- B. The project is ahead of schedule.
- C. Earned value is 245,000 USD.
- D. None of the above.

Answer - B



BAC = 500,00 USD.

CV = -45,000 USD.

AC = 200,000 USD.

PV = 140,000 USD.

$$CV = EV - AC$$

$$-45,000 = EV - 200,000$$

$$EV = 200,000 - 45,000 = \mathbf{155,000 \text{ USD.}}$$

$$SV = EV - PV = 155,000 - 140,000 = \mathbf{15,000 \text{ USD.}}$$

Since the schedule variance is positive, the project is ahead of schedule.

PRACTICE QUESTION

11. You recently took over the responsibilities of a project manager from another person. While reviewing the project cost, you realize there has been an item of \$120,000 waiting to be paid, which, you think, is a disputed item. Your project CPI is 1.1. In this case:

- A. Since this is a small amount compared to the total project value, it can be paid ignoring all objections.
- B. Since the project CPI is greater than one, you do not have any difficulties with the cost of the project. You can, hence, bury the payment in contingency account and move on with the project.
- C. You should inform your manager and ascertain the facts of the matter before making the payment.
- D. You must refuse the payment.

Answer



- **C**– You should inform your manager and ascertain the facts of the matter before making the payment.

As a project manager, you are supposed to deal with any issue or alleged unethical practice and not ignore it.

PRACTICE QUESTION

12. An estimating technique that uses a statistical relationship between historical data and other variables (such as square footage in construction or lines of code in software development) is known as:

- A. Parametric Estimating
- B. Analogous Estimating
- C. Bottom-up Estimating
- D. Historical Analysis

Answer



A – This technique is known as Parametric Estimating. It can produce higher levels of accuracy, depending on the sophistication as well as on the underlying resource quantity and cost data built into the model.

PRACTICE QUESTION

13. While reviewing the performance reports of your project, you notice that the Cost Performance Index (CPI) is 1.2 and Schedule Performance Index (SPI) is 0.8. In this case you should:

- A. Make changes to the performance baseline to improve the SPI.
- B. Evaluate options to crash or fast-track the project
- C. Inform the management why the project got delayed
- D. Tell the team-members that they have no option but to meet the target dates as stated in the project schedule

Answer



- **B** – Since SPI is 0.8, the project is behind schedule. But since CPI is 1.2, you are under budget. Option B (Fast tracking or crashing) is the only alternative that helps solve the problem. They may increase costs but decrease the project timelines.

PRACTICE QUESTION

14. You are the Project Manager and in the process of midway review at the end of the first year of a \$50,000 project. The earned value analysis shows that the PV is \$25,000, the EV is \$20,000 and the AC is \$15,000. What can be determined from these figures?

- A. The project is behind schedule and over budget
- B. The project is ahead of schedule and under budget
- C. The project is ahead of schedule and over budget
- D. The project is behind schedule and under budget

Answer



D – The project is behind schedule and under budget.

PRACTICE QUESTION

15. You are a senior project manager working for RETAMART, a retail shopping network that sells various consumer products. As part of the expansion plan approved by the board of directors, you are a project manager for a new plant. Due to transportation problems, the project has experienced delays; the Schedule Performance Index (SPI) is at 0.6 and the Cost Performance Index (CPI) is at 0.7. However, you expect some improvements over the next few weeks, which may increase the SPI to 1.1 and the CPI to 0.9. Which of the following statements is true if your anticipated changes materialize?

- A. The project is overspent and behind schedule
- B. The project is overspent but ahead of schedule
- C. The project is on schedule and under budget
- D. The project is under budget but behind schedule

Answer



B – The cost performance index below 1 indicates that the project is over budget, and the schedule performance index above 1 indicates that the project is ahead of schedule. If all of your anticipated changes happen to be true, the project will be overspent but ahead of schedule because the schedule performance index will be greater than 1.

PRACTICE QUESTION

16. You have a project with a budget of 500,000 USD. An incident happened during the execution phase that cost you a lot of money. However, you are sure that this will not happen again, and that you can continue with your calculated performance for the rest of the project. To date, you have spent 200,000 USD, and the value of the completed work is 175,000 USD.

Calculate the Estimate at Completion (EAC).

Answer

Estimate at Completion = Money spent to date + (Budgeted cost for all the work – Earned Value)

$$\text{EAC} = \text{AC} + (\text{BAC} - \text{EV})$$

We are given that: Actual Cost (AC) = \$200,000; Budget at Completion (BAC) = 500,000; Earned Value (EV) = 175,000

$$\begin{aligned}\text{Hence, EAC} &= 200,000 + (500,000 - 175,000) \\ &= 200,000 + 325,000 = \mathbf{\$525,000}.\end{aligned}$$